

Pasquale Morrone 2/19/26

PO

CHE 313 Transport Phenomena III
Homework #2 (Due: 2/17/26, 2:30 pm)

TA office hour (T, 11am-1pm; TR, 12:30-2:30 pm): Aelix Acquah (aacquah2@uic.edu), EIB 170

Note: Explain your answers in detail!

1. (2pts) Textbook (Wankat, P. C. 5th edition), 4.D2.
2. (2pts) Textbook (Wankat, P. C. 5th edition), 4.D4.
3. (2pts) Textbook (Wankat, P. C. 5th edition), 4.D6.
4. (2pts) Textbook (Wankat, P. C. 5th edition), 4.D7.
5. (2pts) Textbook (Wankat, P. C. 5th edition), 4.D12.

Problem 4D.2 (2)

$$y = \frac{q}{q-1}x - \frac{z_F}{q-1} \quad \text{Slope: } \frac{1}{q-1}$$

a.) $z_F = 0.60 \quad v/F = 0.37$

$$\frac{0.63}{0.63-1}x - \frac{0.60}{0.63-1}$$

$$q-1 = -0.37$$

$$\text{Slope} = 0.63 / -0.37 = -1.7027$$

$$y_{\text{int}} = 1.6216$$

$$y = -1.7027x + 1.6216$$

b.) $z_F = 0.40 \quad \text{at } 84.1^\circ\text{C}$

$$x = 0.1661 \quad y = 0.5089$$

$$z_F = (1 - v/F)x + (v/F)y$$

$$v/F = \frac{z_F - x}{y - x} = \frac{0.40 - 0.1661}{0.5089 - 0.1661} = 0.6823$$

$$q = 1 - v/F = 1 - 0.6823 = 0.3177$$

$$y = \frac{0.3177}{0.3177-1}x - \frac{0.40}{0.3177-1}$$

$$\text{Slope} = \frac{0.3177}{-0.6823} = -0.4657$$

$$y_{\text{int}} = \frac{-0.40}{-0.6823} = 0.5863$$

$$y = -0.4657x + 0.5863$$

c.) $z_F = 0.40 \quad \text{liquid at } 20^\circ\text{C}$

$$T_b = 80.7^\circ\text{C} \quad \Delta T = 60.7^\circ\text{C}$$

$$q = 1 + \frac{C_{p,L}(T_b - T_F)}{\lambda}$$

$$C_p = 90 \text{ J/molK} \quad \lambda = 39.8 \text{ kJ/mol}$$

$$C_p \Delta T = 90(60.7) = 5460 \text{ J/mol} = 5.46 \text{ kJ/mol}$$

$$q = 1 + 5.46/39.8 = 1.137$$

$$\text{Slope} = 1.137/0.137 = 8.30$$

$$y_{\text{int}} = -2.92$$

$$y = 8.30x - 2.92$$

D.) $Z_F = 0.40$ Vapor at 120°C

$$q = \frac{h_V - h_F}{h_V - h_L} = \frac{C_p(T_F - T_{\text{dew}})}{\lambda}$$

$$C_p \Delta T = 46(31.52) = 1.45 \text{ kJ/mol}$$

$$q = 1.45 / 39.8 = -0.037$$

~~-0.048~~

$$\text{Slope} = -0.037 / -1.037 = 0.0357$$

$$y_{\text{int}} = -0.4 / -1.037 = 0.3857$$

$$\lambda = 39.8 \text{ kJ/mol}$$

$$T_{\text{dew}} = 88.48^\circ\text{C}$$

$$\Delta T = 31.52^\circ\text{C}$$

$$C_p = 46 \text{ J/mol K}$$

$$y = 0.0357x + 0.3857$$

E.) $Z_F = 0.50$ 1 mol Vapor $\rightarrow$ 1/2 mol F

$$\Delta h = \lambda / 12$$

$$q = 1 + \Delta h / \lambda = 1 + 1/12 = 1.0833$$

$$\text{Slope: } \frac{1.0833}{0.0833} = 13$$

$$y_{\text{int}} = \frac{-0.50}{0.0833} = -6$$

$$y = 13x - 6$$

F.) $Z_F = 0.20$ 70% vaporized $V/F = 0.7$ $L/F = 0.3$

$$Z_F = \frac{L}{F}x + \frac{V}{F}y = 0.3x + 0.7y$$

$$q_L = 1.0$$

$$q_V = 0$$

$$x = 0.0426 \quad y = 0.2674$$

Vertical

Horizontal

Problem 4D.4 (2) $x_D = 0.04$ $F_1 = 100 \text{ kmol/h}$ $F_2 = 80 \text{ kmol/h}$
 $V/F = 0.80$ $q_2 = 0.20$

A.) $F_1 + F_2 = D + B \rightarrow 180 = D + B$

$L = F_1 = 100$

$q_2 F_2 = 0.2(80) = 16$

$(1 - q_2)F_2 = 0.8(80) = 64$

$\bar{L} = 100 + 16 = 116$

$V = \bar{V} + 64$

$D = V = \bar{V} + 64$

$B = 180 - D = 180 - (\bar{V} + 64) = 116 - \bar{V}$

$$\frac{\bar{V}}{B} = \frac{\bar{V}}{116 - \bar{V}}$$

$y = \frac{\bar{L}}{\bar{V}}x - \frac{B}{\bar{V}}x_D$ Bottom

$\bar{L} = 116$ $x_B = 0.04$ $(0.04, 0.04)$

$y = \frac{a}{a-1}x - \frac{z_2}{a-1}$ $q = 0.20$
 $z_2 = 0.40$

$y = -0.25x + 0.50$

Table: $x = 0.20 \rightarrow y = 0.417$

$x = 0.25 \rightarrow y = 0.478$

$y = y_D + \frac{100}{B}(x - 0.60)$

$y_{eq}(x) = 0.417 + \frac{0.478 - 0.417}{0.25 - 0.20}(x - 0.20) = 0.417 + 1.22(x - 0.20)$

$0.417 + 1.22(x - 0.2) = -0.25x + 0.50$

$0.173 + 1.22x = -0.25x + 0.50$

$1.47x = 0.327 \rightarrow x = 0.222$

$M_{min} = \frac{0.444 - 0.04}{0.222 - 0.04} = 2.216$

$y = -0.25(0.222) + 0.5 = 0.444$

$m = \frac{\bar{L}}{\bar{V}} = \frac{116}{\bar{V}} \rightarrow \bar{V}_{min} = \frac{116}{M_{min}} = 52.3$

$B_{min} = 116 - \bar{V}_{min} = 63.66$

$\left(\frac{\bar{V}}{B}\right)_{min} = \frac{52.34}{63.66} = 0.822$
 0.8113

$$B.) \quad \bar{V}/B = 1.5 \quad B = 116 - \bar{V} \rightarrow \bar{V} = 1.5(116 - \bar{V})$$

$$\bar{V} = 174 - 1.5\bar{V} \rightarrow 2.5\bar{V} = 174 \rightarrow \bar{V} = 69.6 \text{ kmol/h}$$

$$B = 116 - 69.6 = 46.4 \text{ kmol/h}$$

$$D = \bar{V} + 64 = 69.6 + 64 = 133.6 \text{ kmol/h}$$

$$100(0.60) + 80(0.4) = 60 + 32 = 92$$

$$92 = D y_D + B x_B$$

$$92 = (133.6) y_D + (46.4)(0.04)$$

$$92 = 133.6 y_D + 1.856 \rightarrow 133.6 y_D = 90.144 \rightarrow y_D = 0.675$$

Bottom

$$y = \frac{116}{69.6} x - \frac{46.4}{69.6} (0.04) \quad y = 1.67x - 0.0267$$

Top

$$y = y_D + 100/D (x - 0.60)$$

$$100/D = 100/133.6 = 0.7485$$

$$y = 0.7485x + 0.2256$$

$$1.67x - 0.0267 = 0.7485x + 0.2256 \rightarrow x = 0.2748 \quad y = 0.4313$$

$$y_{cc}(0.04) = 0.1048$$

$$f = \frac{y_D - y_6}{y_7 - y_6} = \frac{0.6747 - 0.6594}{0.7257 - 0.6594} = 0.231$$

$$6 + 0.231 = 6.231 \text{ trays}$$

$$N_{tot} = 1 + 6.231 = 7.231 \rightarrow$$

7 Stages

Problem 4D.6

$$P = 1 \text{ atm} \quad q = 1$$

$$F = 50 \text{ kmol/h} \quad x_D = 0.9 \quad x_B = 0.05$$

$$L/D = 1 \quad \bar{V}/B = 2$$

A.) $L/D = 1 \rightarrow L = D$

$$\bar{V}/B = 2 \rightarrow \bar{V} = 2B$$

$$V = L + D = D + D = 2D$$

$$V' = V$$

$$L' = L + F$$

$$L' = \bar{V} + B = 2B + B = 3B$$

$$L' = L + F = D + 50 = 3B$$

$$2D = 2B \rightarrow D = B$$

$$\text{overall: } F = D + B$$

$$50 = D + B = 2D \rightarrow D = 25 \quad B = 25$$

$$L = D = 25 \quad V = 2D = 50$$

$$L' = L + F = 25 + 50 = 75 \quad V' = V = 50$$

$$y = L/V x + D/V x_D$$

$$y = 0.5x + 0.45$$

$$y = \frac{L'}{V'} x - \frac{B}{V'} x_B \rightarrow$$

$$y = 1.5x - 0.025$$

$$X = Z_F$$

C.) $0.5 Z_F + 0.45 = 1.5 Z_F - 0.025$

$$Z_F = 0.475 \text{ or } 47.5\%$$

B.) $x = 0.70 \rightarrow y = 0.870$

$$(x_D, y_D) = (0.9, 0.9)$$

$$x = 0.80 \rightarrow y = 0.915$$

$$x_1 = 0.70 + \frac{0.90 - 0.87}{0.915 - 0.87} (0.10) = 0.7667$$

$$y_1 = 0.5(0.7667) + 0.45 = 0.833$$

$$x_2 = 0.6 + \frac{0.833 - 0.825}{0.87 - 0.825} (0.10) = 0.6186$$

$$y_2 = 0.5(0.6186) + 0.45 = 0.7593$$

$$x_3 = 0.4 + \frac{0.7593 - 0.729}{0.779 - 0.729} (0.10) = 0.4606$$

$$z_F = 0.475$$

Tray 3 from the top

$$X_4 = 0.3 + \frac{0.6658 - 0.665}{0.729 - 0.665} (0.10) = 0.3013$$

$$y_4 = 1.5(0.3013) - 0.025 = 0.4270$$

$$X_5 = 0.1 + \frac{0.4270 - 0.418}{0.517 - 0.418} (0.05) = 0.1045$$

$$y_5 = 1.5(0.1045) - 0.025 = 0.1318$$

$$X_6 = 0 + \frac{0.1318}{0.134} (0.02) = 0.0197$$

$$F = \frac{X_5 - X_B}{X_5 - X_6} = \frac{0.1045 - 0.05}{0.1045 - 0.0197} = 0.642$$

$$\text{Trays} = 5 + 0.642 = 5.64 + 1 = 6.64 \approx 7 \text{ total Stages}$$

②
Problem 4D.7

$$x_D = 0.9 \quad x_B = 0.13 \quad L/V = 0.8 \\ z_F = 0.3 \quad V/B = 1.0$$

$$y = \frac{L}{V}x + (1 - L/V)x_D \rightarrow y = 0.8x + 0.18$$

$$y = \frac{L}{V}x + \frac{B}{V}x_B \rightarrow y = 2x - x_B = 2x - 0.13$$

$$A.) x_1 = 0.8 + \frac{0.9 - 0.865}{0.929 - 0.865} (0.10) = 0.8547$$

$$y_1 = 0.8(0.8547) + 0.18 = 0.8638$$

$$x_2 = 0.7 + \frac{0.8638 - 0.802}{0.865 - 0.802} (0.10) = 0.7980$$

$$y_2 = 0.8(0.7980) + 0.18 = 0.8184$$

$$x_3 = 0.7 + \frac{0.8184 - 0.802}{0.865 - 0.802} (0.10) = 0.7261$$

$$y_3 = 0.8(0.7261) + 0.18 = 0.7608$$

$$x_4 = 0.6 + \frac{0.7608 - 0.739}{0.802 - 0.739} (0.10) = 0.6347$$

$$y_4 = 0.8(0.6347) + 0.18 = 0.6877$$

$$x_5 = 0.50 + \frac{0.6877 - 0.674}{0.739 - 0.674} (0.10) = \boxed{0.5211}$$

$$B.) y_R = 0.262 + \frac{0.13 - 0.1}{0.05} (0.348 - 0.262) = 0.3136$$

$$y_R = 2x_1 - 0.13 \rightarrow x_1 = \frac{y_R + 0.13}{2} = 0.2218$$

$$y_1 = 0.478 + \frac{0.2218 - 0.2}{0.05} (0.478 - 0.417) = 0.4436$$

$$x_2 = \frac{y_1 + 0.13}{2} = 0.2868$$

$$y_2 = 0.478 + \frac{0.2868 - 0.25}{0.05} (0.524 - 0.478) = \boxed{0.512}$$

$$c.) \quad 0.8x + 0.18 = 2x - 0.13 \rightarrow 0.31 = 1.2x \rightarrow x = 0.2583$$

$$y = 0.8(0.2583) + 0.18 = 0.3867$$

$$(x, y) = (0.2583, 0.3867)$$

$$m_q = \frac{0.3867 - 0.3}{0.2583 - 0.3} = \frac{0.0867}{-0.0417} = -2.08$$

$$\frac{q}{q-1} = -2.08 \rightarrow q = \frac{-m_q}{1-m_q} = \frac{2.08}{1+2.08} = \boxed{0.675}$$

Feed on Tray 8 below the total Condenser

$$N_{total} = 8 + 1 = \boxed{9}$$

$$y_D = 0.70 \quad x_P = 0.987$$

$$x_B = 0.01$$

$$z_S = 0$$

$$\frac{B}{S} = 5 \rightarrow B = 5S$$

Problem 4D.12

$$F = 1000 \quad z_F = 0.20$$

$$D = P + R$$

$$0.70D = 0.987P + x_R R \quad R = D - P$$

$$0.70D = 0.987P + x_R (D - P)$$

$$0.70D = x_R D + (0.987 - x_R)P$$

$$P = \frac{(0.70 - x_R) D}{(0.987 - x_R)}$$

$$F + S + R = B + D$$

$$0.2F + x_R R = 0.01B + 0.70D$$

$$x_R = x_B = 0.01$$

$$A.) 0.2(1000) = 0.01B + 0.987P$$

$$200 = 0.01B + 0.987P$$

$$1000 + S = B + P$$

$$1000 + S = 5S + P \rightarrow P = 1000 - 4S$$

$$200 = 0.01(5S) + 0.987(1000 - 4S)$$

$$200 = 0.05S + 987 - 3.948S$$

$$200 = 987 - 3.898S$$

$$S = 201.90$$

$$B = 5S = 1009.49$$

$$P = 1000 - 4S = 192.41$$

$$0.70D = 0.987P + 0.01R$$

$$0.70D = 0.987P + 0.01(D - P)$$

$$0.70D = 0.977P + 0.01D \rightarrow 0.69D = 0.977P \rightarrow D = \frac{0.977}{0.69} P$$

$$D = 272.44$$

$$R = D - P = 272.44 - 192.41 = 80.03$$

$$B.) \quad x = 0.5732 \quad y = 0.6841$$

$$x = 0.6763 \quad y = 0.7385$$

$$x_0 = 0.5732 + \frac{0.7 - 0.6841}{0.7385 - 0.6841} (0.6763 - 0.5732) = \boxed{0.603}$$

$$y = L_0/V_1 x + D/V_1 y_D \quad V_1 = L_0 + D = 2D$$

$$V_1 = 2(272.44) = 544.88 \rightarrow L_0/V_1 = D/V_1 = 0.5$$

$$\boxed{y = 0.5x + 0.35}$$

Optimum recycle feed tray: Bottom Tray

$$y = z_F = 0.20$$